

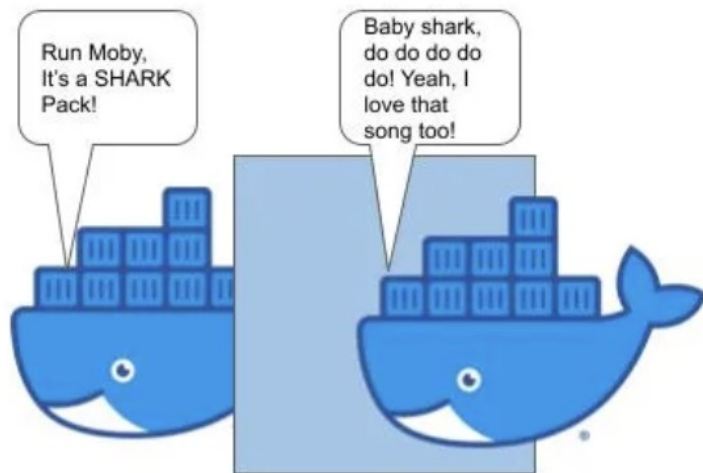
Kubernetes + ML Deployment

Recap

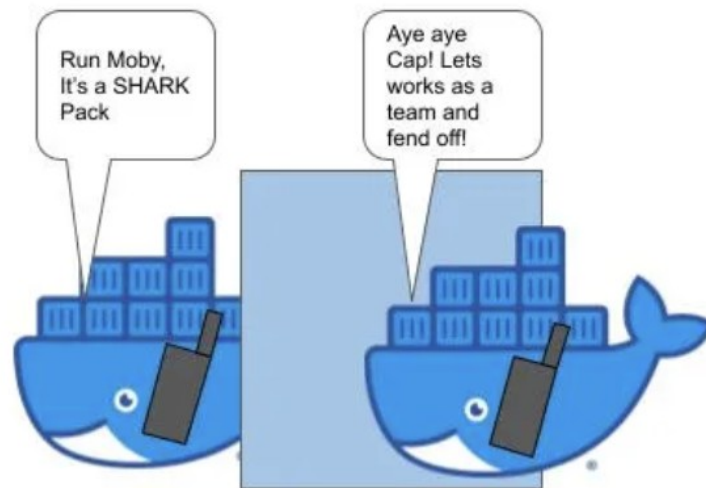
- Cloud
- Virtualization
- Containers
- Docker
- Docker Images
- Docker Networking
- Kubernetes

```
docker tag ml-app:v1 quay.io/ppnaik/ml-app:v1
```

```
docker push quay.io/myorg/ml-app:v1
```



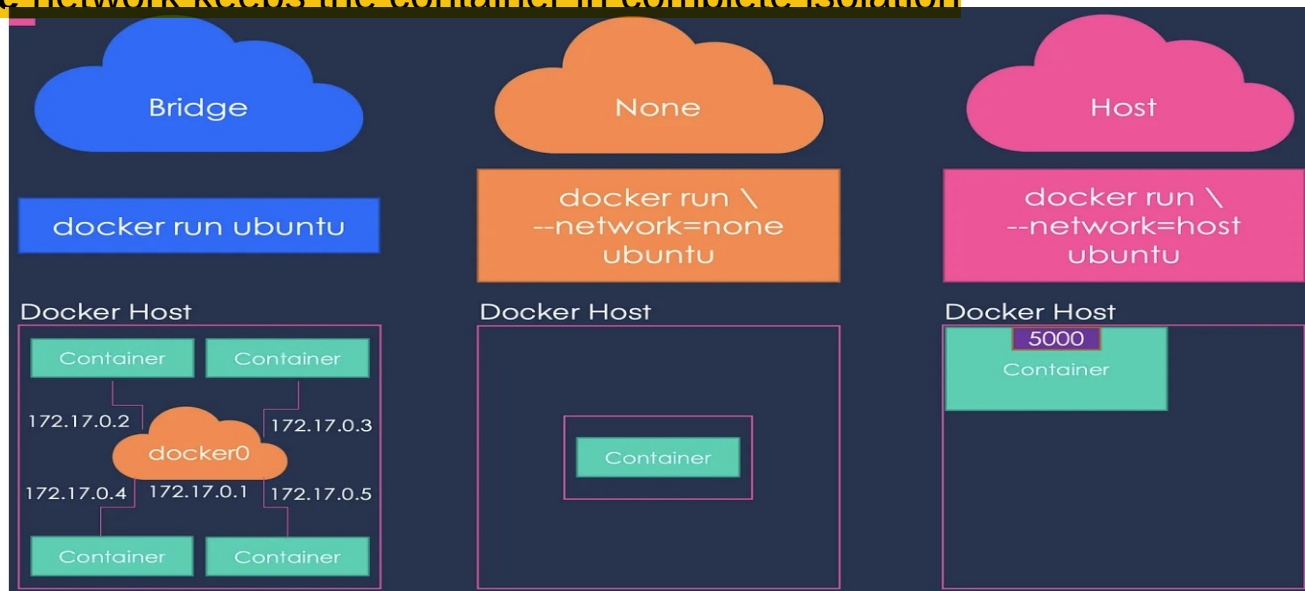
Barrier to communication between docker containers



Communications using the Network Bridge

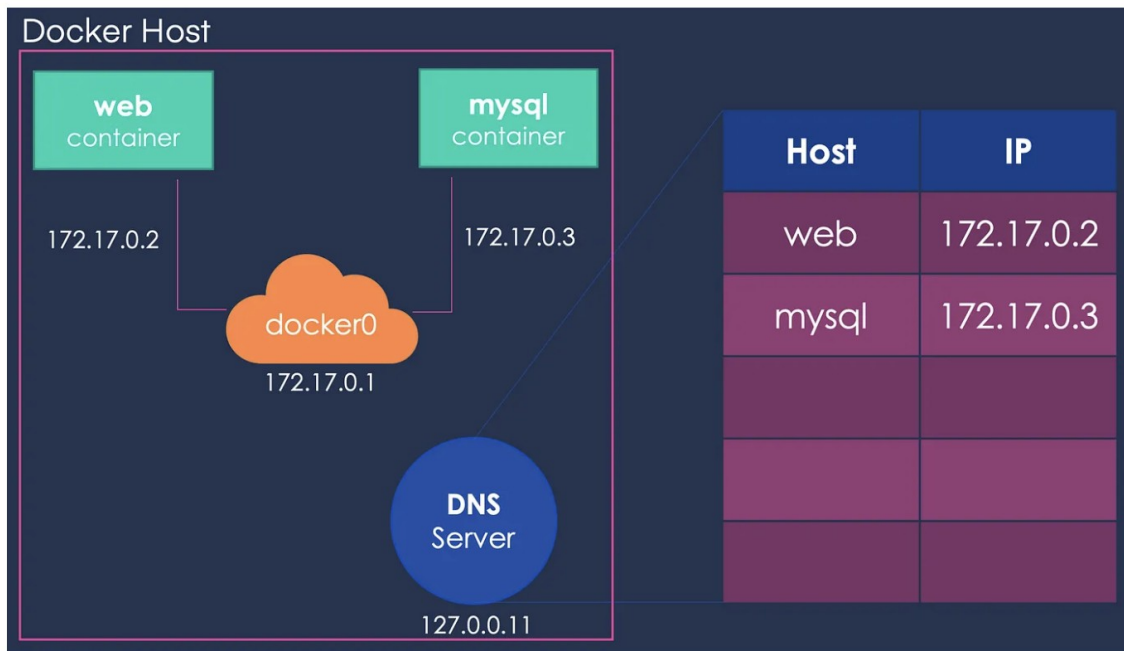
Docker Networking

- Three networks created automatically - Bridge, Host, and None
- The **Bridge** network assigns IPs in the range of 172.17.x.x to the containers
- Host Mode: Accessible on the same port on the docker host without any explicit port mapping
- **None** network keeps the container in complete isolation

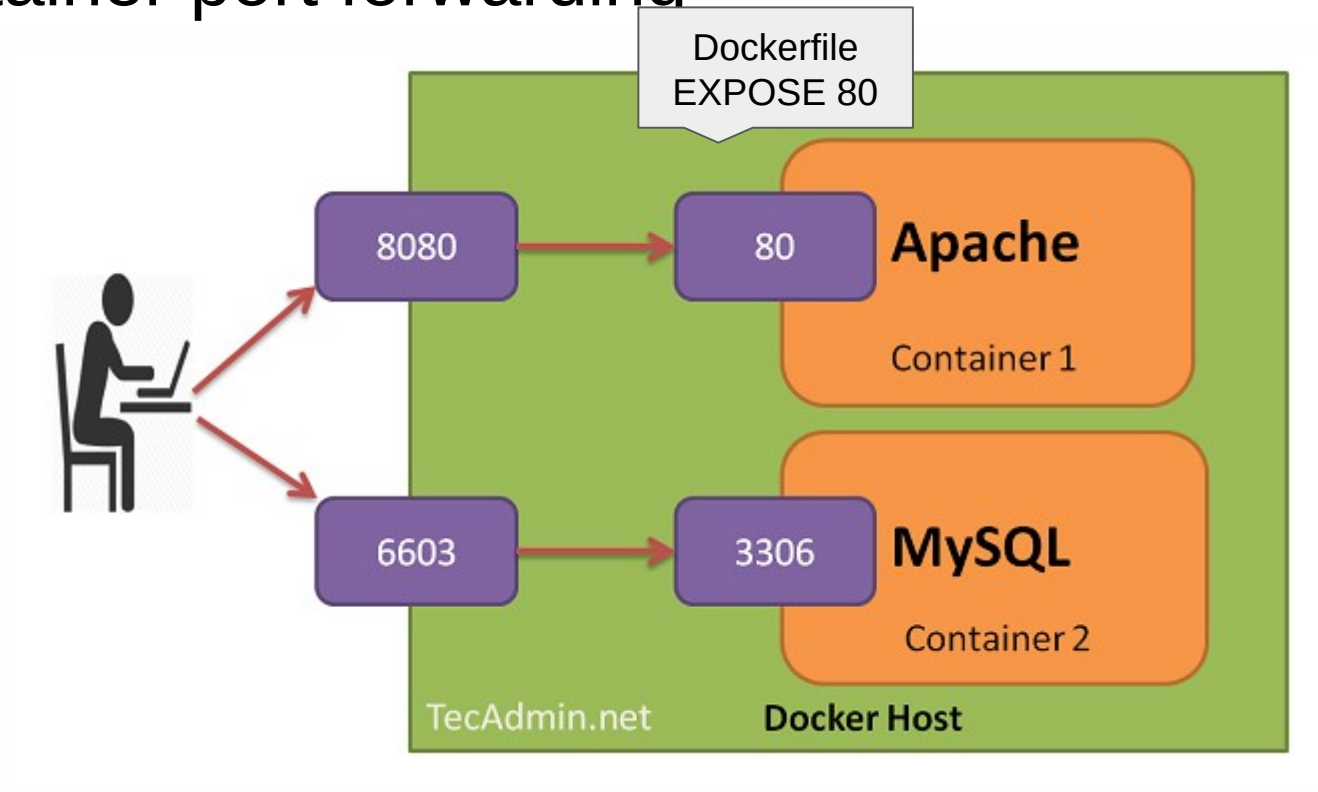


Docker Bridge Network

- Default network container connect to bridge



Container port forwarding

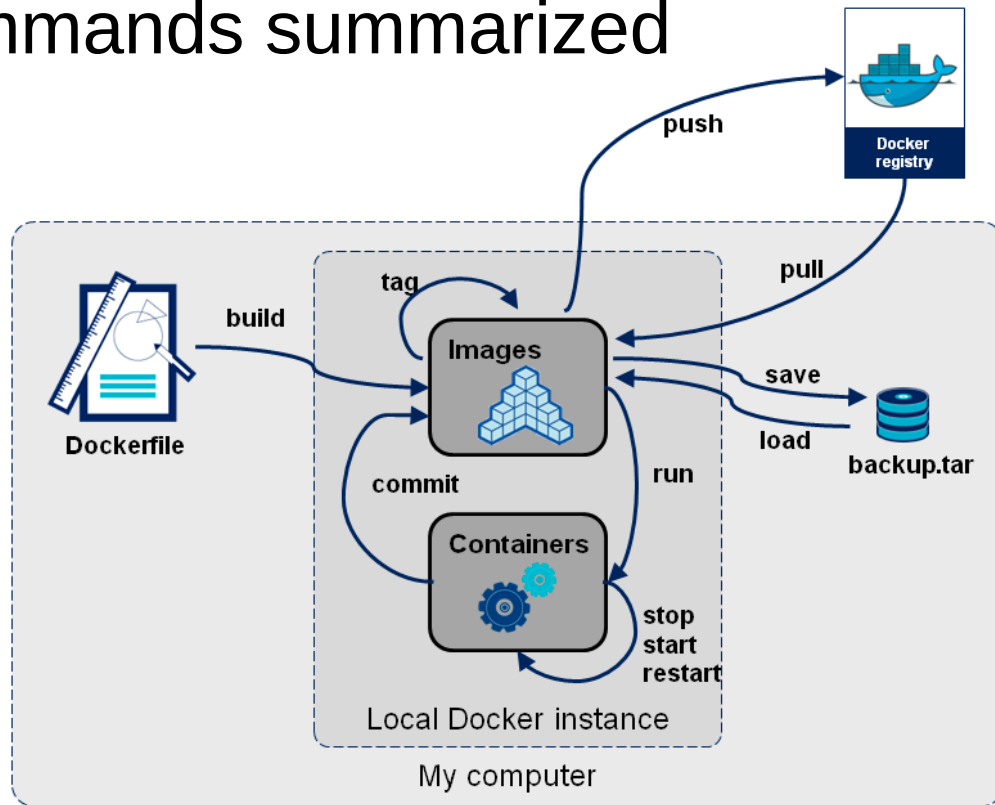


Port forwarding

```
docker run -d --name mycontainer1 -p 8080:80 mypython-app:v1
```

See in action: go to browser and enter URL localhost:8080

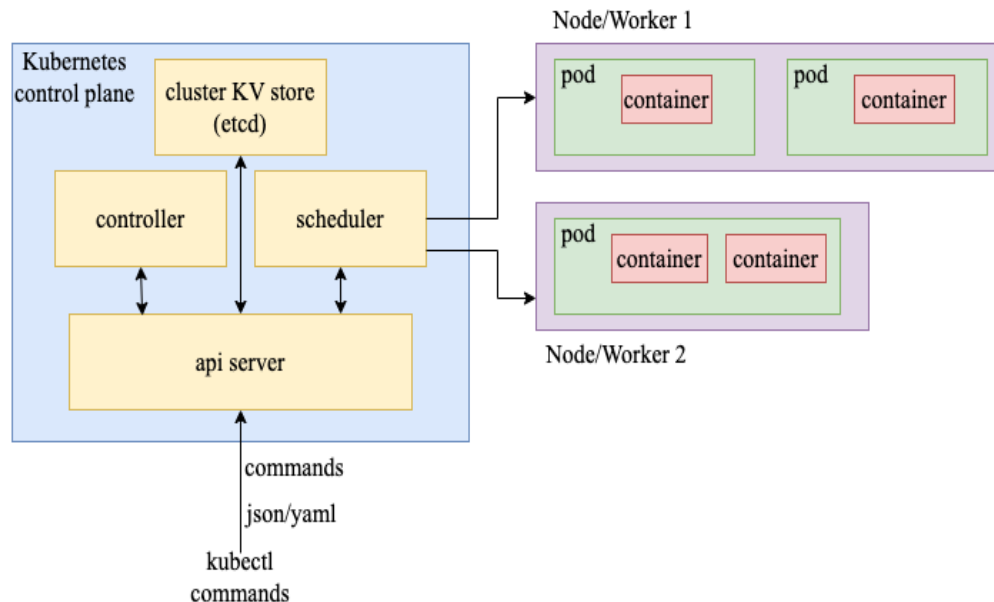
Docker commands summarized



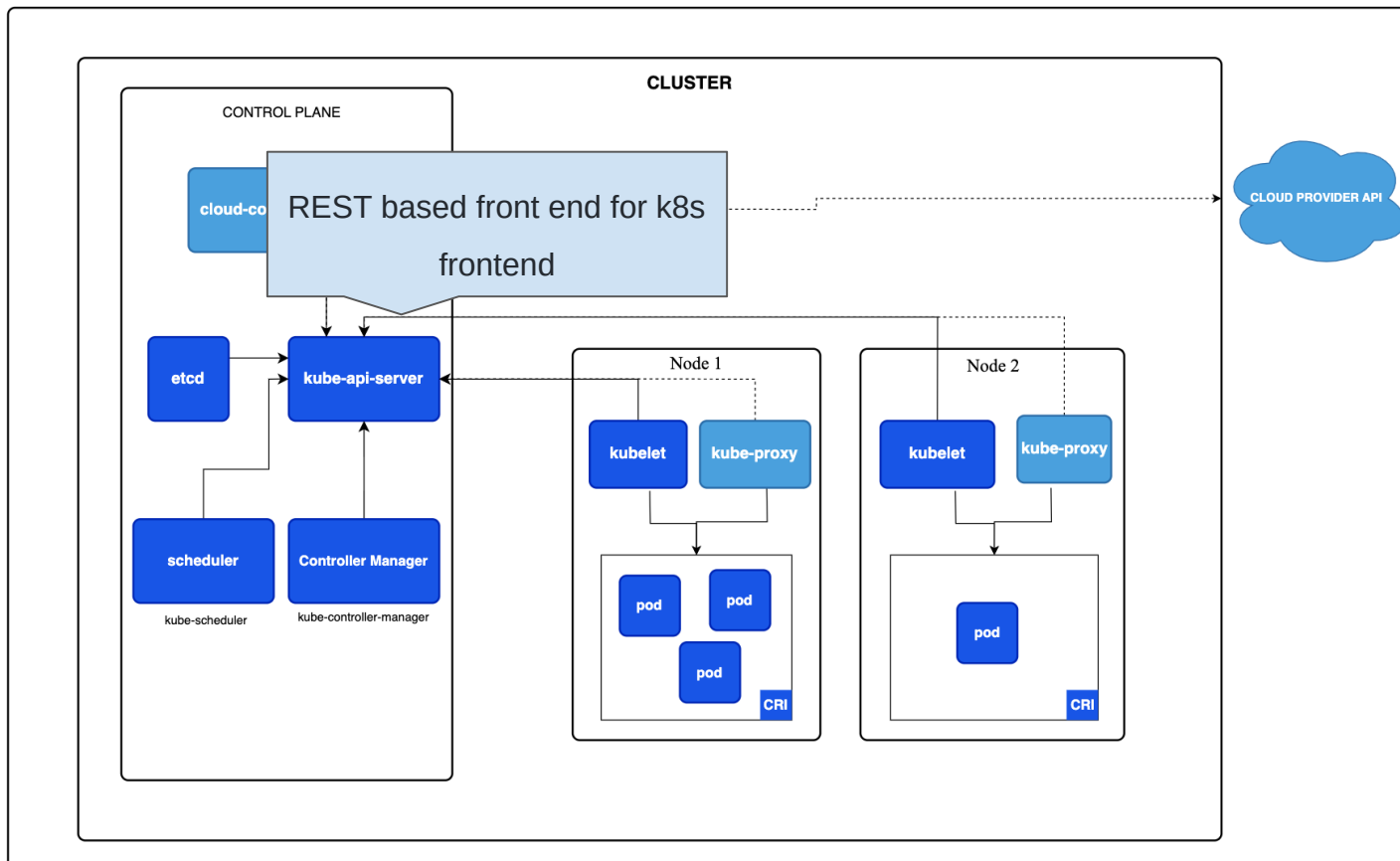
Container orchestration

What is kubernetes?

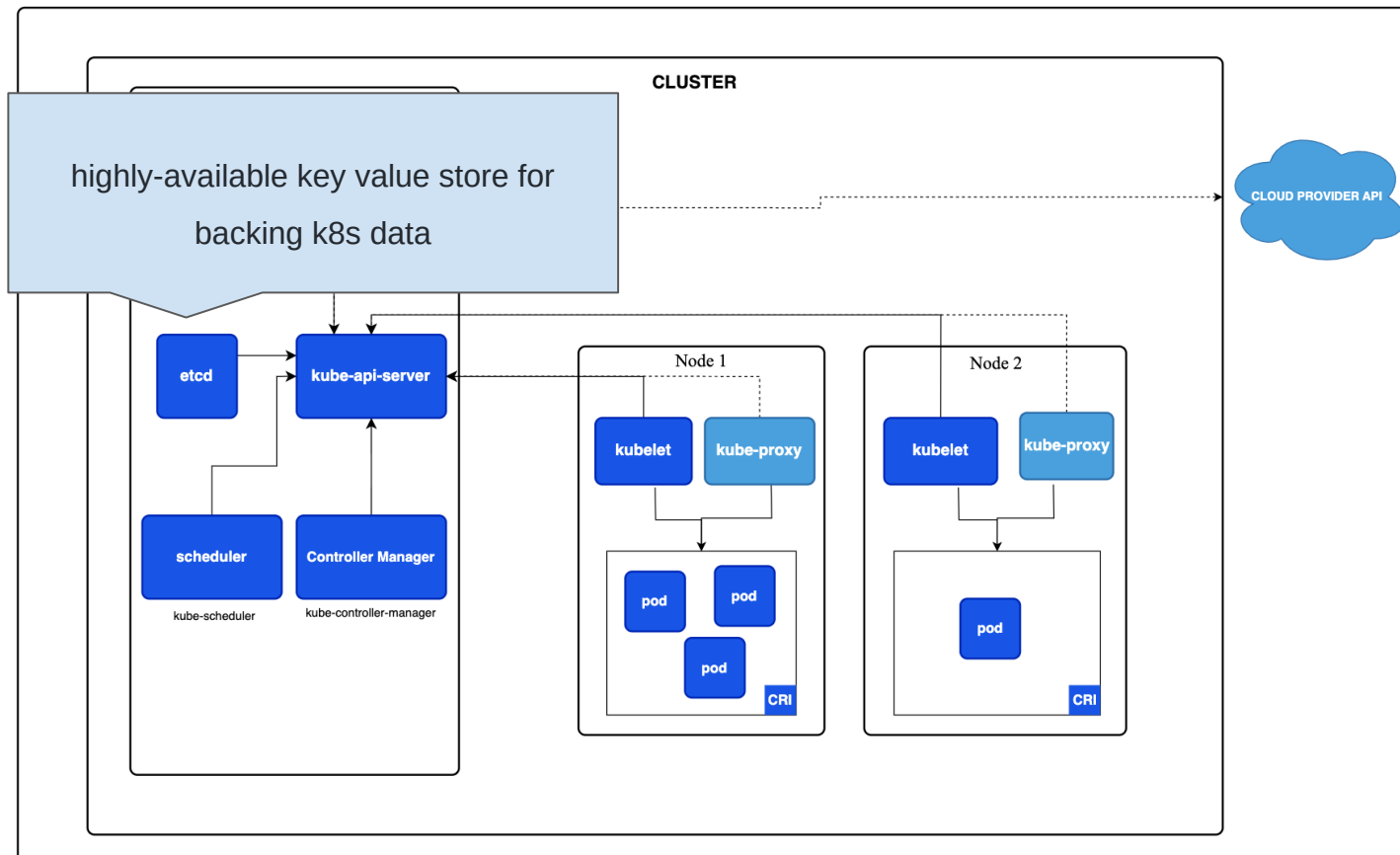
- Docker: Single machine container deployment
- Kubernetes (k8s): Container Orchestration
 - Across a cluster of machines
 - Manage automated deployment, scaling



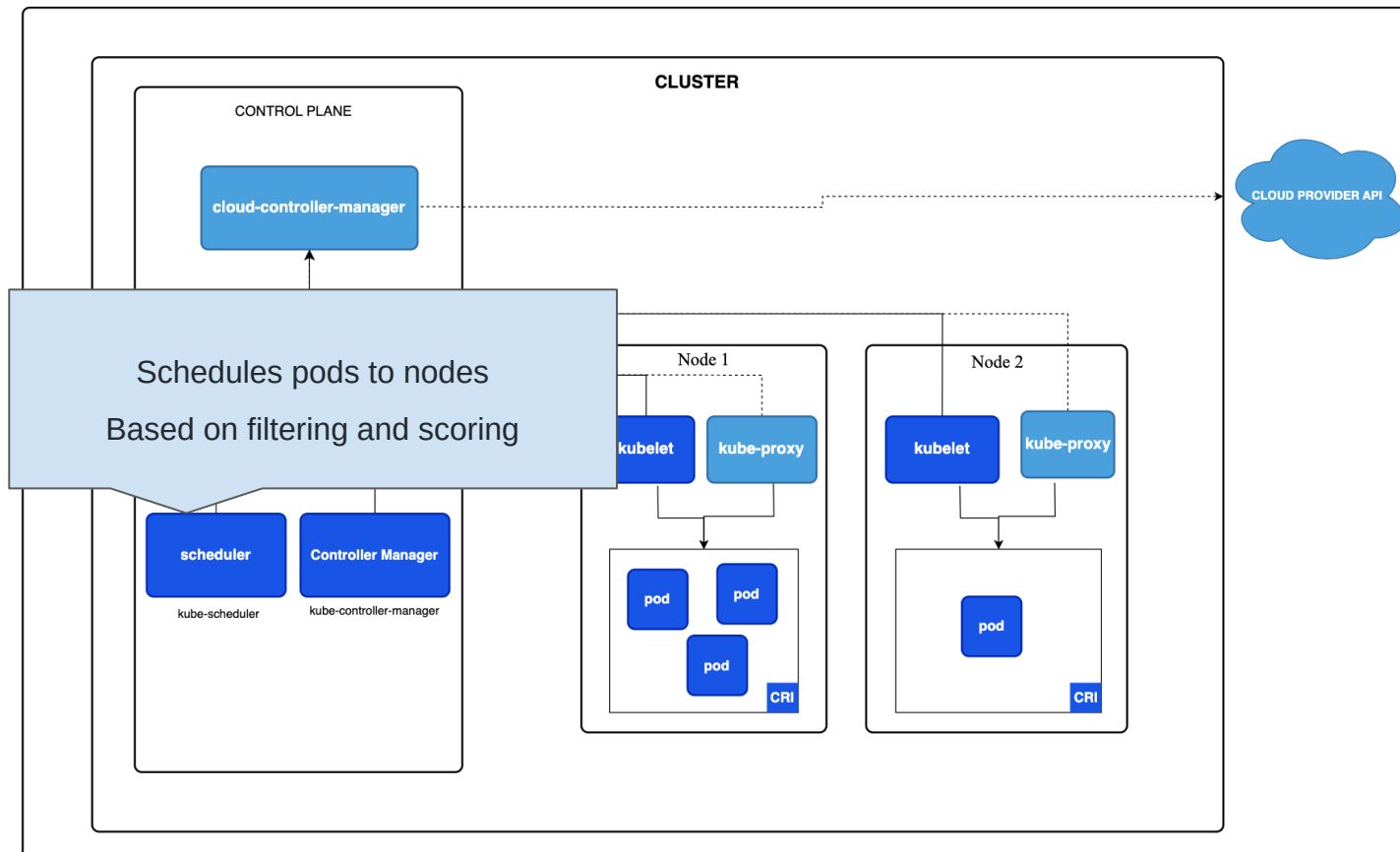
Kubernetes Architecture



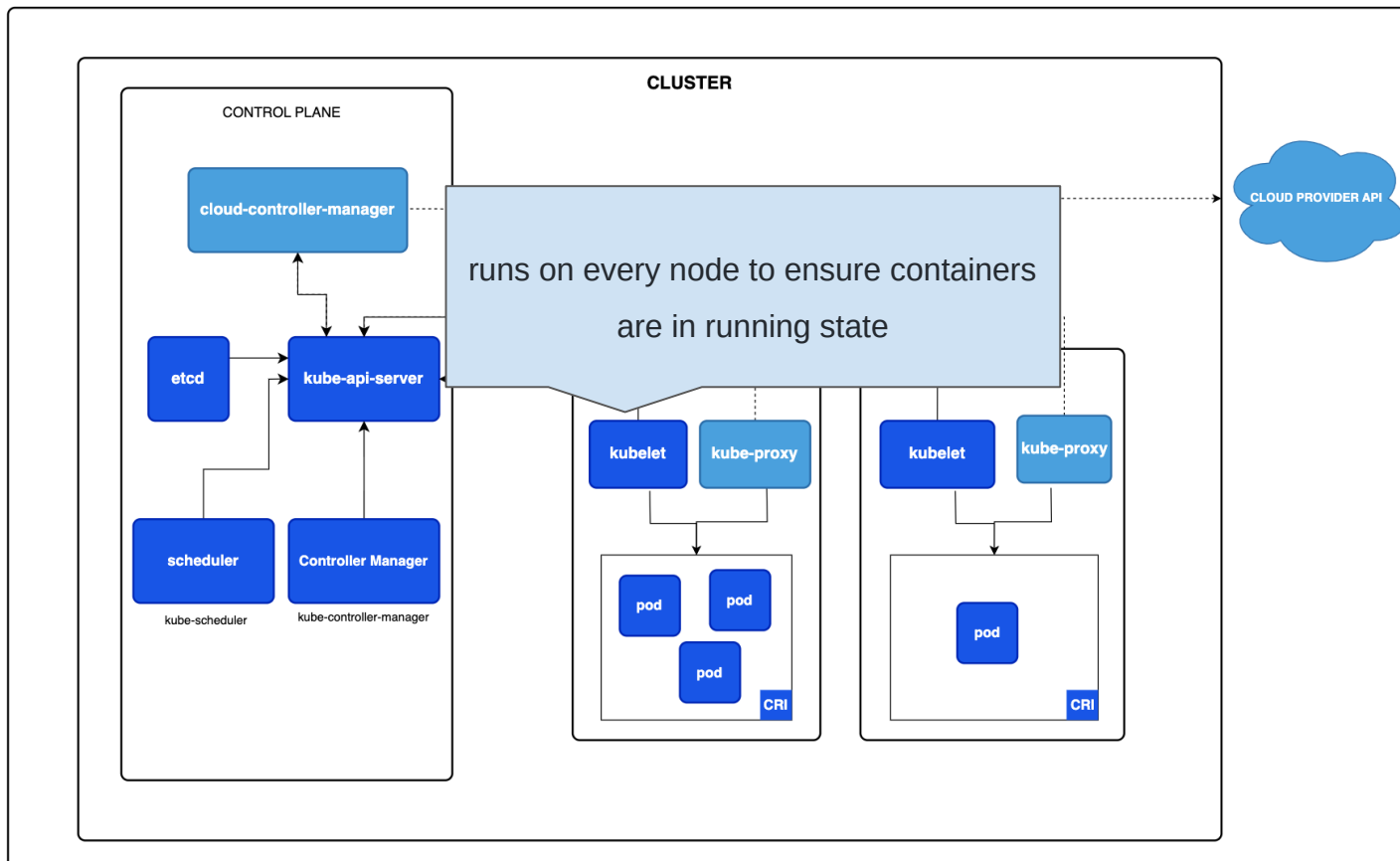
Kubernetes Architecture



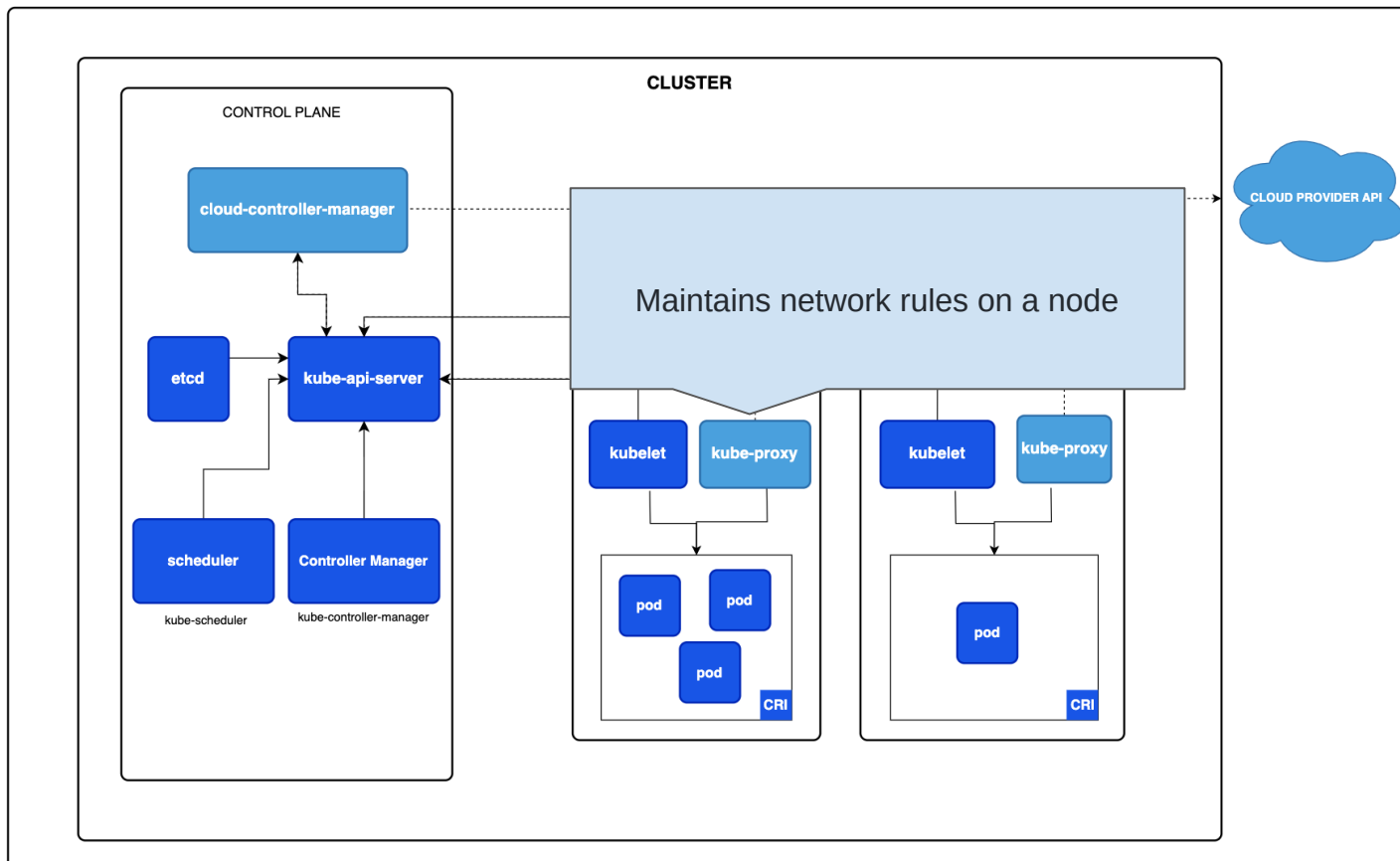
Kubernetes Architecture



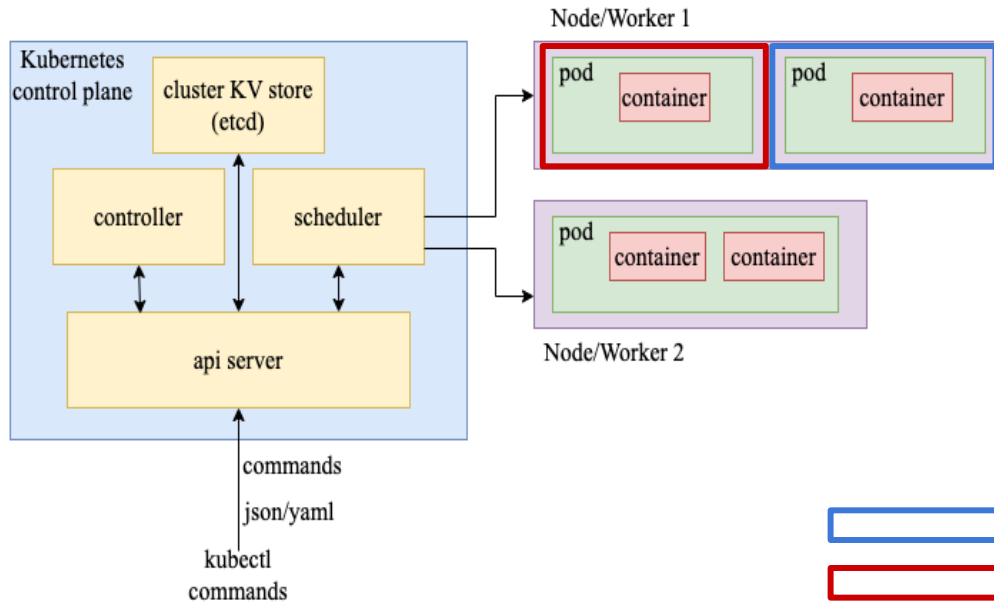
Kubernetes Architecture



Kubernetes Architecture



Namespace



Namespace: Isolated environment for each user

Kubectl get started

- Create a namespace
- `kubectl create namespace <urname>`
- Go to the namespace:
- `kubectl config set-context --current --`

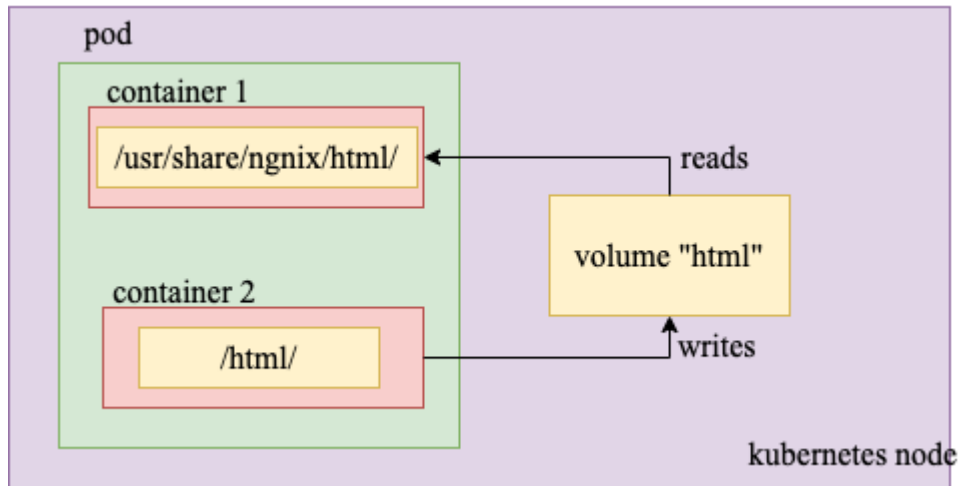
To test locally:

- `namespace=<namespace>`
- Install minikube
- Install kubectl
- Install docker/podman
- `minikube start --driver=docker`
- `minikube kubectl create namespace abc`
- `minikube kubectl -- config set-context --current -- namespace=abc`
- `alias kubectl="minikube kubectl --"`

Pods

Application specific logical host.

- single unit of deployment
- group of containers with shared storage and network resources.



Pod Example add namespace

- Create a single nginx pod:

- `kubectl apply -f nginx-pod.yaml -n <namespace>`

- Check the pod:

- `kubectl describe pod nginx`

- `Kubectl exec -it nginx -- /bin/ba`

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
spec:
  containers:
    - name: nginx
      image: nginx
      ports:
        - containerPort: 80
```

Another Pod Yaml Example

```
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: netshoot
5  spec:
6    containers:
7      - name: netshoot
8        image: nicolaka/netshoot
9        command: ["/bin/bash"]
10       args: ["-c", "while true; do ping 172.30.227.43; sleep 60;done"]
```

Deployments

- Kubernetes Controllers
- Features:
 - Rolling Update
 - Rollback
 - Version Control



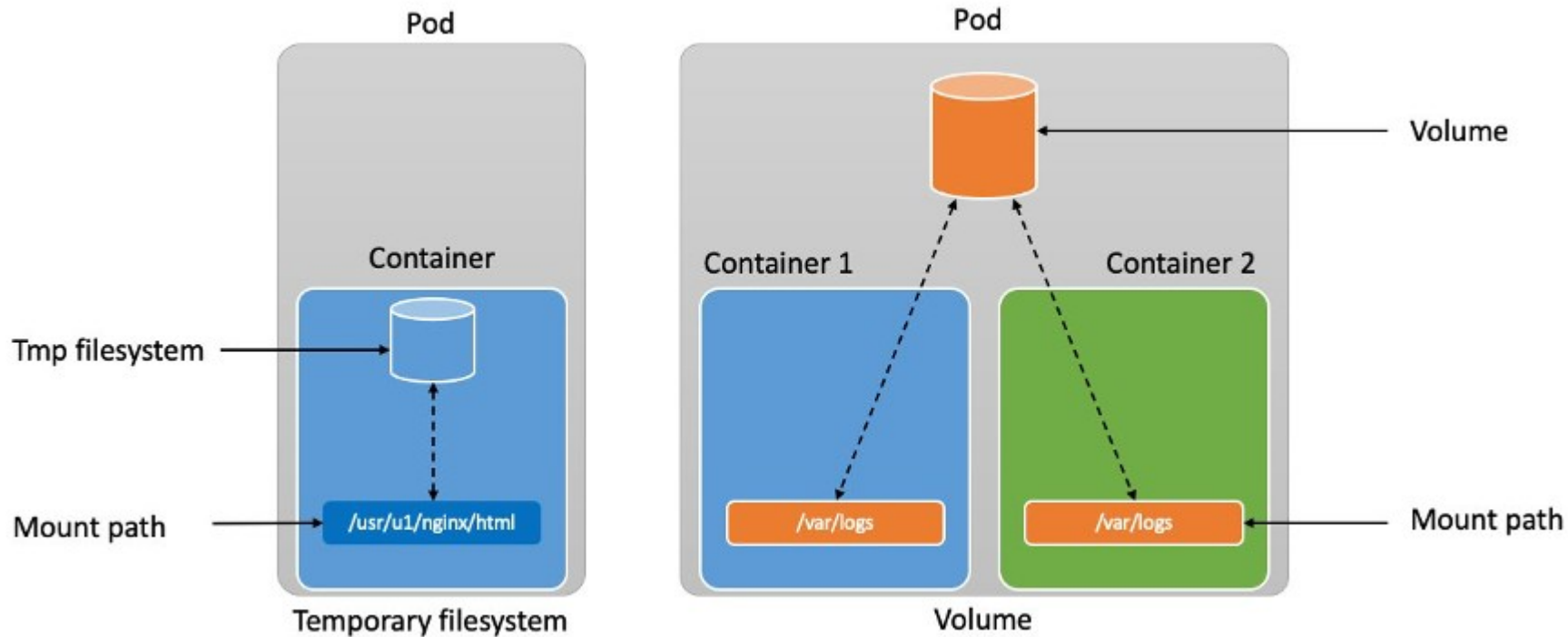
```
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: nginx-deployment
5    labels:
6      app: nginx
7  spec:
8    replicas: 3
9    selector:
10     matchLabels:
11       app: nginx
12   template:
13     metadata:
14       labels:
15         app: nginx
16     spec:
17       containers:
18         - name: nginx
19           image: nginx
20           ports:
21             - containerPort: 80
```

Deployment Example

- `kubectl apply -f nginx_dep.yaml`
- Replicas, delete and see them start



Persistent Volume



Persistent Volume and PersistentVolumeClaim

- PersistentVolume (PV):
 - is a cluster resource
 - Implemented as plugin (hostpath,local,nfs)
 - Have a lifecycle independent of a pod
- PersistentVolumeClaim (PVC) :
 - is a storage request from a user.
 - Request for volume similar to CPU, memory
 - PVC access modes: ReadWriteOnce, ReadOnlyMany, ReadWriteMany, or ReadWriteOncePod

Statefulset and Daemonset

Statefulset

- Deployment and scaling of stateful pods
- Stateful pod: requires persistent storage

Demonset

- An instance of the pod runs on each node
- System daemons and background processes

Services

- Abstraction for an application running multiple replica
- Service discovery to find matching endpoint
- Multi-port Service
- Type:
 - ClusterIP:
 - NodePort:
 - LoadBalancer:
 - ExternalName:
- Service Discovery

Types of Service

- ClusterIP:
 - Default service type. Accessible within the cluster
- NodePort:
 - Allocates a port on each node to forward request to the service
 - `--service-node-port-range` flag (default: 30000-32767)
- Load Balancer
 - Traffic from the external load balancer is directed at the backend Pods.
 - You can specify a `loadBalancerIP` or use ephemeral IP
- ExternalName
 - Service to a DNS name
 - `my.database.example.com`

Service

- `kubectl apply -f nginx-service.yaml`
- `kubectl describe service nginx-service`
- Check connectivity (use the netshoot pod)
 - `curl <serviceIP:port>`
 - Which IP, which port ??

```
1  apiVersion: v1
2
3  kind: Service
4
5  metadata:
6
7    name: nginx-service
8
9  spec:
10
11    selector:
12
13      app: nginx
14
15    ports:
16
17      - port: 8765
18
19        targetPort: 80
20
21    type: LoadBalancer
```

ML-app example

```
kubectl apply -f configmap.yaml
```

```
kubectl apply -f pv.yaml
```

```
kubectl apply -f deployment.yaml
```

```
kubectl port-forward -n ml-app svc/ml-app 8002:80
```

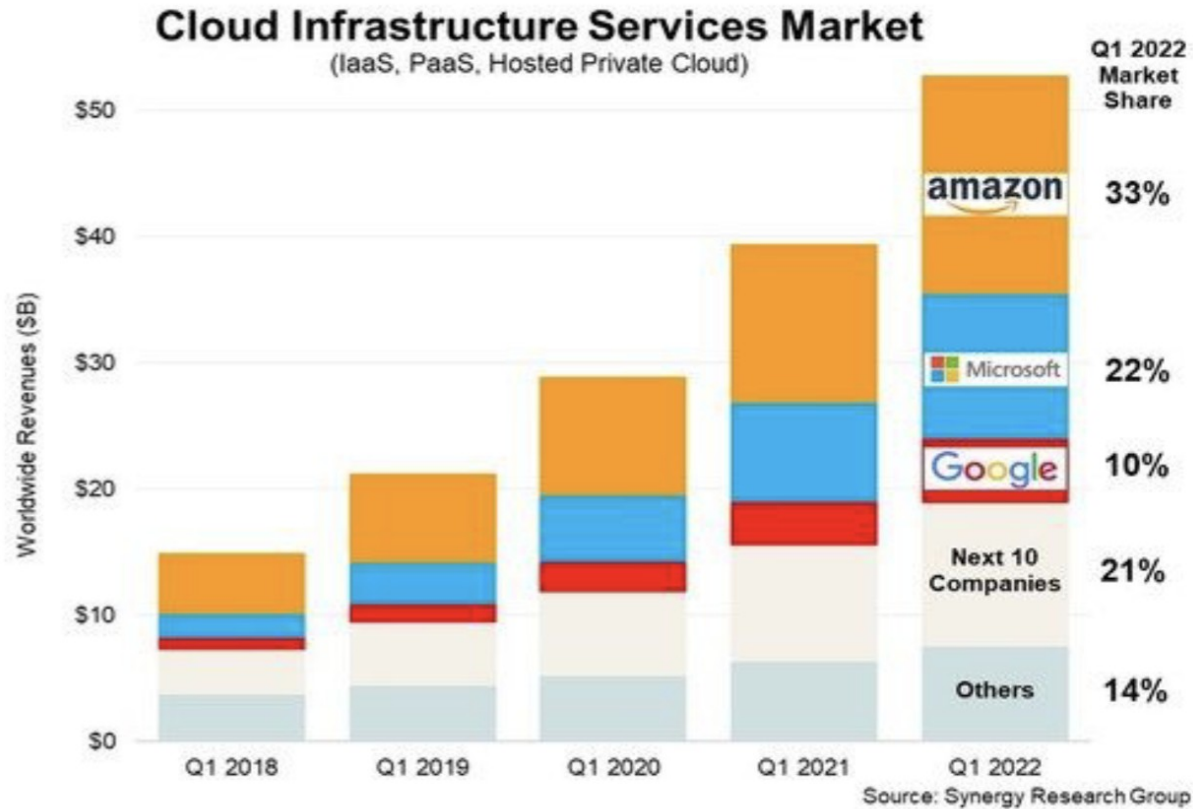
```
curl -X POST http://localhost:8002/predict \  
-H "Content-Type: application/json" \  
-d '{"texts": ["This movie was amazing!"]}'
```

Try it out

https://github.com/pnl-iiitd/acm_fsn/tree/main (k8s and service)

<https://medium.com/aspnetrun/deploying-microservices-on-kubernetes-35296d369fdb>

Cloud Market



Hyperscale Cloud Providers

	AWS EC2	Azure	GCP
Providers	Amazon	Microsoft	Google
Strengths	Market leader/ Pricing Model	Integration with Microsoft suite	Fastest growing for AI-ML market
Users	Netflix, Airbnb, Coursera	HSBC, Starbucks, HP	Paypal, Unilever, Toyota
Scale	200 Data centers (22 regions)	More regions (66)	26 regions, Low latency

Hybrid Cloud

- Public Cloud
- Private Cloud/ On Prem



Multi-Cloud

